

Learning Curves

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Overfitting

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Read both curves for Learning curves and variability. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Learning curves and variability. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/ranges and a careful comparison.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Training sizes 20,40,80 yield validation accuracies 62%,70%,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

6. Solve the given case

With 20,40,80 training leaves, fixed-set validation accuracy is 60%,70%,78%.

Interpret the curve and uncertainty.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

As training examples grow from 50 to 100 to 200, validation accuracy is 60%, 72%, 78%. What does the curve suggest, and what does it not guarantee?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 28 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Plot Plant Classifier train/validation performance at several training sizes and document the chosen next action.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Training sizes 20,40,80 yield validation accuracies 62%,70%,73%. Is doubling again guaranteed to yield 76%? Describe a defensible next experiment.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Summarize repeats

Calculate the mean and range for the 20-row condition.

My reasoning and evidence

13. W2 Read both curves

Why can training error rise while validation error falls?

My reasoning and evidence

14. W3 Avoid extrapolation

Can the 40-row result guarantee a mean MAE of 2.5 at 80 rows?

My reasoning and evidence

15. W4 Apply independently

At sizes 12 and 24, validation errors are [6, 8, 7] and [4, 7, 7]. Calculate means/ranges and a careful comparison.

My reasoning and evidence

16. W5 Explain the spread

Why is an observed range not a confidence interval?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Learning Curves

Read a learning curve with repeated-run spread rather than drawing a certainty arrow.

At 10 training plants validation errors across 3 runs are [6, 5, 7]; at 20 plants errors [4, 5, 3], on the same 10-case validation set.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Learning Curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Summarise first size

Mean 6 errors; range 5-7, corresponding to mean 40% accuracy.



2. Summarise larger size

Mean 4 errors; range 3-5, corresponding to mean 60% accuracy.



3. State bounded pattern

More training plants improved the average in these supplied runs; variability remains.

CHANGE ONE THING

A 30-plant experiment gives errors [4, 4, 4].

Explain the changed result using the matching lesson.

What comparison is valid if the validation set changes between sizes? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

